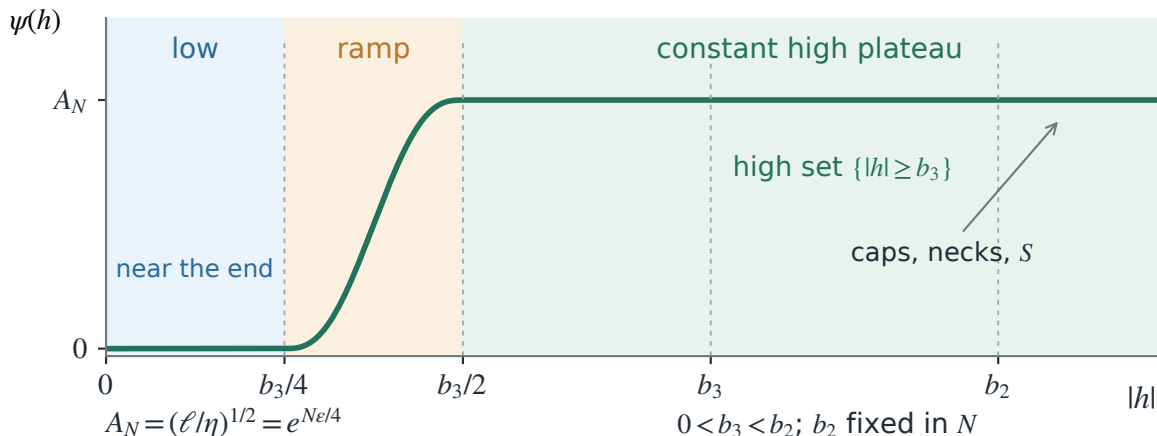


Height cutoff and the exclusion of obstacle contact

Height values only; horizontal scale schematic



$$\mathcal{N}_{2s_N}^{\text{bg}}(O_N) \subset \{\psi = A_N\}$$

Metric-neighborhood containment, not a width on the height axis.

End mass: vanishing upper error

$$M_N \leq e^\epsilon (E_* + a_N)$$

$$a_N \leq P_N(\ell + \ell^{1/2}) \longrightarrow 0$$

The cutoff leaves end energy unchanged.

Contact: prohibitive area cost

$$P_{\tilde{g}} \geq e^{2A_N - P_N} > e^{P_N}$$

$$\text{Fixed enclosing sphere: } P_{\tilde{g}} \leq e^{P_N}$$

No obstacle contact for large N .